



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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CAPSTONE PRESENTATION

“Design and Development of a Biomedical Equipment Maintenance Management System Using Python”

COURSE :PYTHON PROGRAMMING

COURSE CODE : CSA0802

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Introduction



- Biomedical equipment is essential for accurate diagnosis, treatment, and patient monitoring in healthcare facilities.
- Proper maintenance of medical equipment ensures patient safety, equipment reliability, and compliance with healthcare standards.
- Many hospitals still use manual methods to record maintenance activities, making it difficult to track equipment history and maintenance schedules.
- Manual record-keeping can lead to delayed servicing, increased equipment downtime, and inaccurate maintenance records.



Problem Statement



- Hospitals and healthcare facilities use a large number of biomedical equipment that require regular maintenance and servicing.
- Many healthcare organizations still rely on manual or paper-based maintenance records, which are time-consuming and prone to errors.
- Manual maintenance systems make it difficult to track equipment history, maintenance schedules, and repair records.
- Delayed preventive maintenance can lead to unexpected equipment failures, increased downtime, and higher maintenance costs.

Literature Review

- Several studies have highlighted the importance of effective maintenance management for biomedical equipment to ensure patient safety and reliable healthcare services.
- Researchers have reported that manual maintenance systems are inefficient, time-consuming, and prone to human errors, leading to poor equipment management.
- Computerized Maintenance Management Systems (CMMS) have been introduced to automate equipment tracking, preventive maintenance scheduling, and maintenance record management.
- Existing maintenance management systems improve equipment availability, reduce downtime, and increase maintenance efficiency through digital record-keeping.

Architecture / Configuration





Engineer to Excel



Project Diagram

Design and Development of Biomedical Equipment Maintenance Management System Using Python





Module 1



User Login and Authentication :

- Allows authorized users to log in securely using a username and password.
- Verifies user credentials before granting access.
- Protects biomedical equipment and maintenance data from unauthorized access.
- Redirects authenticated users to the system dashboard.



Module 2

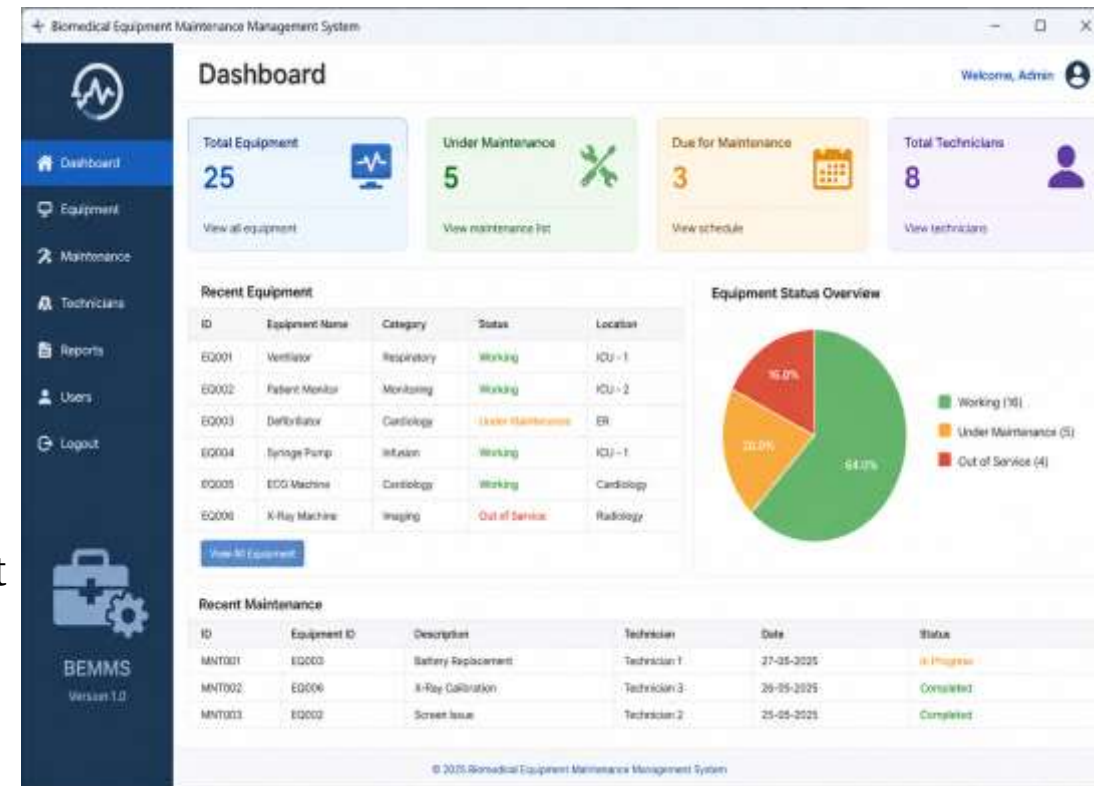


Equipment Registration :

- Stores details of biomedical equipment such as name, model, serial number, and location.
- Assigns a unique ID to each equipment item.
- Updates and manages equipment records efficiently.
- Maintains a centralized database for easy tracking and maintenance.

Result

- The Biomedical Equipment Maintenance Management System was successfully developed using Python.
- The system effectively manages biomedical equipment records and maintenance schedules.
- It stores repair history and generates maintenance reports accurately.
- The system reduces manual paperwork and minimizes equipment downtime.
- It improves maintenance efficiency and provides a user-friendly interface.



conclusion

- The Biomedical Equipment Maintenance Management System using Python was successfully designed and developed.
- The system simplifies equipment registration, maintenance scheduling, and repair record management.
- It reduces manual work, improves accuracy, and minimizes equipment downtime.
- The project provides a cost-effective, user-friendly, and efficient solution for healthcare facilities to manage biomedical equipment maintenance.



Applications



- Used in hospitals to manage biomedical equipment maintenance.
- Helps biomedical engineers track equipment service and repair history.
- Schedules preventive maintenance to reduce equipment downtime.
- Maintains a centralized database of medical equipment records.
- Generates maintenance reports for easy monitoring and documentation.



Future Scope



- Integrate cloud storage for secure access and backup of maintenance records.
- Develop a mobile application for real-time maintenance updates and notifications.
- Add IoT support to monitor the status of biomedical equipment remotely.
- Implement AI-based predictive maintenance to detect faults before equipment failure.
- Include barcode/QR code scanning for faster equipment identification and tracking.

References

- **World Health Organization (WHO).** *Inventory and Maintenance Management Information Systems for Medical Devices*, 2025.
Source: <https://www.who.int/publications/i/item/9789240111257>
- **World Health Organization (WHO).** *Computerized Maintenance Management System*, WHO Medical Device Technical Series, 2012.
Source: <https://www.who.int/publications/i/item/9789241501415>
- **Python Software Foundation.** *Python Documentation*.
Source: <https://docs.python.org/3/>
- **International Organization for Standardization (ISO).** *ISO 13485:2016 – Medical Devices: Quality Management Systems*.
Source: <https://www.iso.org/iso-13485-medical-devices.html>



THANK YOU